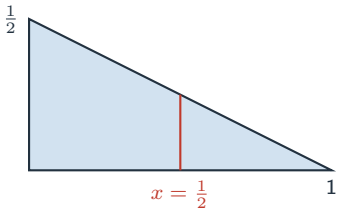
Условие

Нека $f_{X,Y}(x,y) = cx + y$ при $x, y \geq 0$, $x + 2y \leq 1$. Намерете c, $\mathbb{E}[Y|X = \frac{1}{2}]$ и плътността на $Z = X + 2Y$.

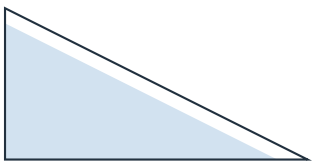
Носителят и c



$1 = \int_0^1 \int_0^{(1-x)/2} (cx + y)dydx = \frac{c}{12} + \frac{1}{24}$, значи $c = \frac{23}{2}$.
При $x = \frac{1}{2}$, $y \in (0, \frac{1}{4})$ с условна плътност, пропорционална на $\frac{23}{4} + y$:

$$\mathbb{E}[Y|X = \tfrac{1}{2}] = \frac{\int_0^{1/4} y(\frac{23}{4} + y)dy}{\int_0^{1/4} (\frac{23}{4} + y)dy} \approx 0.126.$$

Плътност на Z



$$F_Z(z) = \int_0^z \int_0^{(z-x)/2} \left(\tfrac{23}{2}x + y\right) dy dx = z^3$$

Отговор. $c = \frac{23}{2}$; $\mathbb{E}[Y|X = \frac{1}{2}] \approx 0.126$; $f_Z(z) = 3z^2 \mathbb{1}_{(0,1)}(z)$.